

Surgiguid e



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Surgery can feel like navigating
uncharted waters...

- ❑ So many instructions...
- ❑ Pre-op, post-op—where do I start?
- ❑ How do I stay on track?



Surgery Can Be a Lot to Handle
Good Thing Surgi Has Eight Arms!

The Problem: “Health Literacy In Surgery”

More than **\$238 billion** is lost each year due to poor health literacy overall

Affects nearly **9 out of 10** adults

Poor health literacy during preoperative and postoperative periods are associated with **disparities**

Lower health literacy is correlated with **worse surgical outcomes**



1. “Talking Points About Health Literacy.” Centers for Disease Control and Prevention, 2021.
2. “Impact of low health literacy on patients’ health outcomes: a multicenter cohort study,” BMC Health Services Research, 2022.
3. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4504415/#CR6>
4. <https://www.sciencedirect.com/science/article/pii/S0022480423002974>

What are the **unmet health literacy needs of patients undergoing surgery** ?

Kathryn Bregna, MD

General Surgery Resident, Rutgers

“[Patients seek] information that clearly explains **indications and contraindications for surgery** , a complete, **specialty-specific guide** for a surgery that includes information on **perioperative care** would be valuable for patients”

Sharon Gonzalez, MD

Interventional Radiology, Rutgers NJMS

It's not very common to have perioperative care information readily available for **emergency procedures**

Patients often seek clear, reliable information about the **medications** they'll need before and after surgery

Our Solution

Help track
post-op
recovery

Organize
pre-
surgery
info

Improve
health
literacy

Improve
patient-
physician
relationship



For patients SurgiGuide can...

Empower
patients

Reduce
post-op
adverse
events

Lower pre-
surgery
stress &
anxiety

Provide
patient-
centered
tools

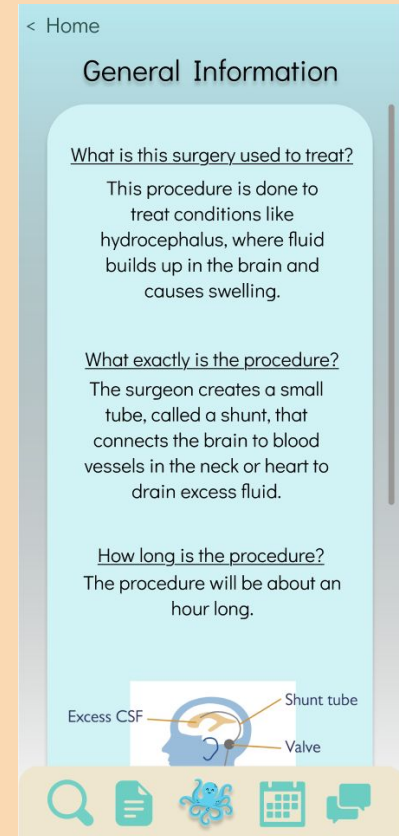
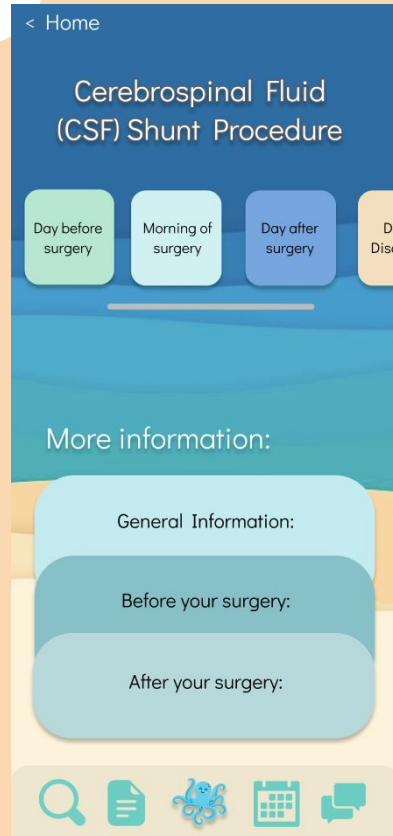


Example of Surgery “Timeline” and General Info:

Peri-operative care for a CSF Shunt placement procedure

Surgeon's Action

The procedure will be about an hour long. The surgeon creates a small tube, called a shunt, that connects the brain to blood vessels in the neck or heart to drain excess fluid. This procedure is done to treat conditions like hydrocephalus, where fluid builds up in the brain and causes swelling. The surgeon may use a tiny camera, called an endoscope, to help guide the procedure.



A Guide to SurgiGuide



Search

Search procedures by speciality, procedure type



General Info

Descriptive diagrams, steps to the surgery, and indications for surgery



Timeline

Pre- and post-operative expectations at different time points



Calendar

View and add events, receive notifications for upcoming appointments



Notes

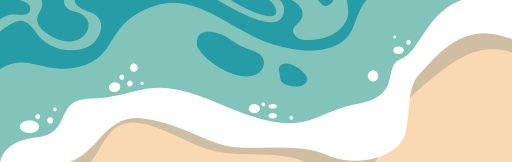
Record thoughts, questions to ask physician

Demo



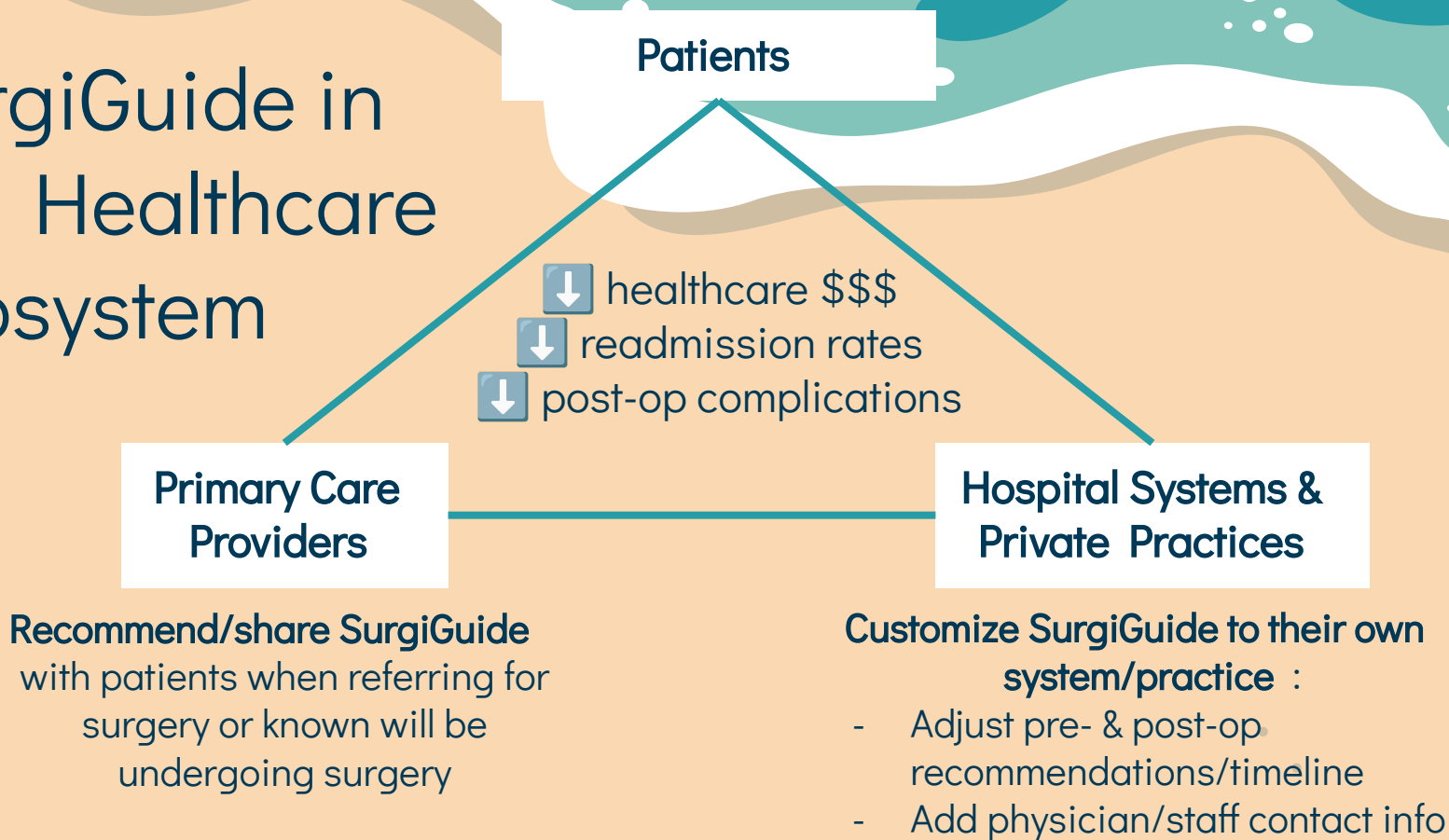


How does our vision for SurgiGuide
compare to the competition?



	SurgiGuide	PostOp	MySurgery	Pip
Pre-Surgery Guidance	Yes	No	Yes	Yes
Post-Surgery Guidance	Yes	Yes	Yes	Yes
Symptom Logging/Notes Section	Yes	No	No	Yes
Health Literacy Enhancements - Surgery explanation, Easy to navigate timeline of surgery	Yes	No	No	Yes
Ease of Information Access for Family members	Yes	No (only patients registered can use)	Yes	Limited
Will work directly with hospital/clinic	Yes	Yes** Limited to Kettering General Hospital (UK only)	No	Yes** Limited to UPMC Presbyterian, UPMC Shadyside, UPMC Horizon, University of Chicago

SurgiGuide in the Healthcare Ecosystem



Next Steps for SurgiGuide:

Partner With Hospital or Clinic

- Collaborate with hospital or surgery center to validate app content
- Gain feedback from residents, surgeons, patients
- EHR integration/ HIPAA compliance

User Testing & Improvements

- User testing: patients, physicians, hospital staff
- Gather feedback on usability, accessibility, and effectiveness in improving health literacy

ChatBot Features & Physician Portal

- Improve chatbot usability and credibility by using AI models and inputting credible medical data
- Create a physician portal for doctors to directly input information to relay to patients

Improve Database

- Add to existing data for more surgeries, specialties, subspecialties
- Make data more accessible: multiple languages, patient populations